

Data Profiling & Initial Observations Report

Dataset Overview: Sales & Transactions Dataset | Total Rows: 1,014 | Total Columns: 17

1. Column Names and Data Types

Column Name	Non-Null Count	Dtype	Column Name	Non-Null Count	Dtype
Invoice ID	1014	str	Sales	994	str
Branch	1014	str	Date	1002	str
City	1014	str	Time	1014	str
Customer type	1014	str	Payment	1014	str
Gender	1014	str	cogs	1014	float64
Product line	1014	str	gross margin %	1014	float64
Unit price	990	float64	gross income	1014	float64
Quantity	1014	int64	Rating	1014	float64
Tax 5%	1014	float64	-	-	-

2 & 3. Data Integrity: Missing Values and Duplicates

Issue Metric	Affected Column / Scope	Count	Percentage
Missing Values	Unit price	24	2.37%
Missing Values	Sales	20	1.97%
Missing Values	Date	12	1.18%
Duplicate Rows	Entire Row Duplication	14	1.38%
Duplicate IDs	Invoice ID (1,000 unique out of 1,014)	14	1.38%

4. Basic Statistical Summary

Metric	Mean	Std	Min	25%	50%	75%	Max
Unit price	55.52	26.33	10.08	33.20	54.89	77.67	99.96
Quantity	5.50	2.93	1.00	3.00	5.00	8.00	10.00
Tax 5%	15.34	11.69	0.51	5.86	12.07	22.43	49.65
cogs	306.74	233.72	10.17	117.29	241.32	448.56	993.00
gross margin %	4.76	0.00	4.76	4.76	4.76	4.76	4.76
gross income	15.34	11.69	0.51	5.86	12.07	22.43	49.65
Rating	6.96	1.72	4.00	5.50	6.90	8.48	10.00

5. Unnecessary & Suspicious Data Identified

- **Incorrect Data Types:** The Sales column is currently formatted as string (object) rather than numeric (float).
- **Inconsistent Text Formatting:** Column Payment has redundant categories (e.g., 'Ewallet' and 'E-wallet').
- **Leading Whitespaces in Column Names:** Columns like ' Customer type' and ' cogs' contain irregular space padding.

- **Constant Value Column:** gross margin percentage has a standard deviation of 0.00 (fixed at ~4.7619%), making it non-informative.
- **Exact Identical Columns:** Tax 5% and gross income share identical distribution metrics (mean, min, max).

6. Summary Observations & Cleaning Recommendations

1. **Duplicate Removal:** Drop the 14 fully duplicated rows to restore proper primary key uniqueness for Invoice ID.
2. **Missing Values Strategy:** Impute missing Unit price values using Sales / Quantity, or vice versa, where applicable.
3. **Type Casting & Standardization:** Cast Sales to float and parse Date / Time into standard datetime formats.
4. **Text Cleaning:** Strip leading spaces from all header titles and harmonize category labels in Payment.